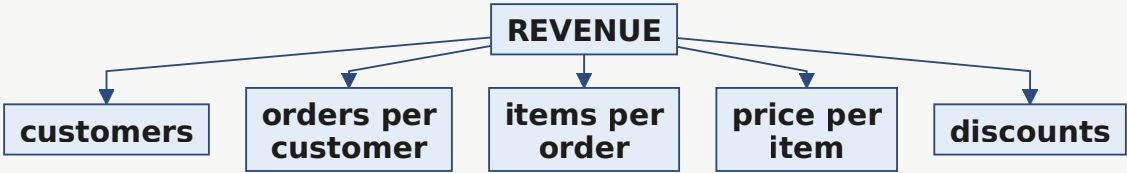


1 The picture, and the one rule



Crux. A rate with no denominator is a rumour, so every branch is written as a numerator over a denominator before any number is computed.

2 Every branch is a metric

Branch	Over what	Costs
Customers	The window, matched both sides	Marketing, slowest
Orders per customer	Distinct customers, same window	Retention, a tier
Items per order	Orders	Merchandising
Price per item	Items	Pricing, risks volume
Discounts	Gross before discount	Margin, directly

3 Mean or median

Use	When
Median	Describing a typical order
Mean	Dividing a total that must reconcile
Both	The first pass on any file

Day 1: mean Rs 18,160, median Rs 2,205. One corporate order carries 88 percent of the revenue.

Crux. When the mean sits far above the median, the gap is the finding and the mean describes nobody.

4 The three lines every leaf is made of

```
total = 0
for order in ORDERS:
    total = total + int(order["amount"])
```

Start at zero. Walk the list. Add. Count adds one, sum adds the value, distinct uses a `set`.

5 Errors you meet today

Message	First move
<code>TypeError: ... 'int' and 'str'</code>	Find the row, then write a rule rather than a patch
<code>KeyError: 'discount'</code>	Ask whether it is missing or absent by design
<code>NameError: name ... not defined</code>	Restart the kernel and run all cells

6 Four readings of one word

Gross bookings. Delivered revenue. Net of returns. Cash collected. Four different numbers live inside the word sales.

Crux. Say which reading you used, every time, or the room spends an hour arguing about the wrong number.

7 The sentence shape

Claim, with its number and its denominator. The caveat that would change it. What you would do next.

That shape closes every day this week, and it opens every good interview answer.

THE WORDS ON THIS SHEET

Revenue tree Revenue written as a product of branches, each of which a team owns
Branch One factor in that product, expressed as a metric

Denominator What a rate is divided by, and the thing that has to match across periods
Accumulator A variable that starts at zero and is added to inside a loop

Record One row as a dictionary of named fields
Median The middle value once the values are sorted

Mean The total divided by the count
Kernel The process holding everything your cells have created so far